

✓ 1. Find USB Insertion / Removal Timestamps

USB activity is stored in the Windows Registry.

📌 Location 1 — USB Device History

HKLM\SYSTEM\CurrentControlSet\Enum\USBSTOR

Inside each device folder you will find:

- *LastWrite* time → **USB insertion timestamp**
- Device serial number
- Vendor ID (VID)
- Product ID (PID)

📌 Location 2 — Mounted Devices (Drive Letter Assigned)

HKLM\SYSTEM\MountedDevices

Shows:

- USB drives mounted
- Drive letter given (E:, F:)
- Time they were mounted

📌 Location 3 — USB Connection Timestamps

HKLM\SYSTEM\CurrentControlSet\Enum\USB

Shows all USB devices ever connected including:

- Keyboard
- Mouse
- Pendrive
- Phone
- External HDD

✓ Tools that show this automatically:

- **Autopsy** → Recent Activity → USB Devices
- **USBDeview** (NirSoft)
- **Magnet AXIOM** → Artifacts → USB Devices
- **Kape** + SBEC modules (SANS Blueprint)

✓ 2. Extract Device ID & Manufacturer

Use the registry path:

HKLM\SYSTEM\CurrentControlSet\Enum\USBSTOR\<DeviceName>\<SerialNumber>\

You will find:

- **Device name** (SanDisk, Kingston, WD)
- **Manufacturer**
- **Serial number**
- **VID** (Vendor ID)
- **PID** (Product ID)

Example entry:

USBSTOR\Disk&Ven_SanDisk&Prod_Cruzer_Blade\1234ABC5678

This tells you:

- Vendor → SanDisk
- Product → Cruzer Blade
- Serial → 1234ABC5678

✓ **Fast Method (GUI):**

Use **USBDeview** (NirSoft).

It shows:

- Device Name
- Serial Number
- First Connect Time
- Last Disconnect Time
- VID/PID

✓ **3. Detect Files Copied to USB (If Detectable)**

Windows does **not** directly store “files copied to USB” logs.

But we detect it using the following artifacts:

A. ShellBags (Folder Open History)

Shows which folders were opened by the user.

Location:

NTUSER.DAT → ShellBags

This can show:

- If the user opened a folder **inside a USB drive**
- Example: “E:\Projects\Confidential\”

Tools:

- ShellBags Explorer (Eric Zimmerman)

- Autopsy → "Shellbags"

B. LNK Files (Shortcut Files)

Windows creates shortcut files for **recently opened files**.

Location:

C:\Users\<User>\AppData\Roaming\Microsoft\Windows\Recent\

If you see:

file.txt.lnk → Target: E:\Data\file.txt

= It confirms the file existed on the USB drive.

This works for:

- Opened files
- Copied files (sometimes triggered)

C. Jump Lists

Windows uses jump lists for recently accessed files.

Location:

C:\Users\<User>\AppData\Roaming\Microsoft\Windows\Recent\AutomaticDestinations\

Jump list entries pointing to E:\ or F:\ = USB interaction.

Tools:

- JLECmd (Eric Zimmerman)
- Autopsy → Recent Activity

D. \$LogFile & \$MFT Artifacts

Using a forensic tool (Autopsy/FTK), you can view:

- MFT entries pointing to E:\ or F:\
- File copy operations (metadata only)
- Timestamps for file creation events on USB

★ BEST TOOLS FOR FILE COPY DETECTION

Tool	What It Shows
Autopsy	All events, shellbags, jump lists, LNK
Magnet AXIOM	USB timeline + file interaction details
X-Ways Forensics	Low level \$MFT analysis

Eric Zimmerman Tools Jump list & shellbag evidence

USBDeview Device details only

✅ **4. Detect Suspicious Data Exfiltration**

Look for:

✔ **A. Large data transfer events**

Using \$MFT "size changed" timestamps:

- If many files created on USB in a short time → data copy activity.

✔ **B. Prefetch Evidence**

Check prefetch for:

explorer.exe

robocopy.exe

xcopy.exe

powershell.exe with copy commands

These indicate mass copying.

✔ **C. Event Logs (Only on servers usually)**

Microsoft-Windows-DriverFrameworks-UserMode/Operational

Shows USB events (connect/disconnect)

🔥 **QUICK PRACTICAL STEPS (Simplified)**

Step 1 — Extract registry hives

- SYSTEM
- SOFTWARE
- NTUSER.dat
- UsrClass.dat

Step 2 — Run Eric Zimmerman Tools

USBDeview

ShellBags Explorer

JLECmd

LECcmd

MFTECmd

Step 3 — Load image into Autopsy

- View "Recent Activity"
- USB Devices

- Shellbags
- LNK files
- Jump lists
- \$MFT timeline

Step 4 — Build timeline

Look for:

- USB plug-in time
- Folders opened on USB
- Files created on USB
- Suspicious copy tools opened